

Louis Kotze

Senior Full-Stack Engineer · Linux Kernel Contributor

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SUMMARY

Senior engineer with 10+ years across full-stack product delivery, platform architecture, and systems-level open-source. Ships production code for enterprise clients in pharmaceuticals, automotive, and FinTech, and contributes upstream to the Linux kernel, wpa_supplicant, NetworkManager, and wireless-regdb — 17 merged commits across five projects. Leads small distributed teams across time zones.

EXPERIENCE

Lead Developer, iOCO Digital · May 2025 – Present

Consulting engagements for enterprise clients. Led a team of 3 engineers delivering a data-visualization platform used across a global pharmaceutical group's internal analytics teams: part of a 14-team product group, coordinating with 4 of those teams plus chapter leads in India, business stakeholders in Switzerland, and a product owner in the US. Currently delivering on a large-scale agricultural traceability platform through UAT.

- Shipped 5 new chart types on the analytics platform (geographic, work-management, and process-insight views), widening its coverage across supply-chain, site operations, and work-management data
- Designed a multi-view layer that lets analysts pivot between renderings of the same dataset without leaving the page
- Drove AI-assisted development adoption within the team: KT sessions, restructuring the codebase for agent-friendly context (Copilot / Claude Code / Codex), and automated Copilot-driven PR reviews
- Cut worst-case report latency from minutes to seconds across a production dataset of hundreds of thousands of members and millions of records: root-caused non-sargable joins and query-planner miscosting with EXPLAIN (ANALYZE, BUFFERS), then fixed with targeted indexes, materialized CTEs, and denormalization — largest single win two orders of magnitude
- Built a bug-to-fix regression corpus (180+ pairs) and cross-environment schema-parity gates, catching defects and schema drift before release
- Cut managed-database running cost by right-sizing RDS storage from a 90-day CloudWatch throttling analysis
- Stacks: Angular 21, Highcharts, GraphQL, Storybook, GitHub Actions; currently React, C#/F.NET, PostgreSQL, AWS, Azure DevOps

Senior Software Developer, BMW IT Hub South Africa · Aug 2021 – May 2025

Led engineering on the Group Document Management System and Technical Action Database for BMW, working across teams in Germany, South Africa, and Portugal (team size 3–7).

- Engineered BMW's Group Document Management System for international legal-document retention: per-country classification drove statutory retention windows, version history preserved historical drafts as required by law, end-of-life prompts flagged documents nearing expiry, and daily compliance jobs purged documents past their statutory due date
- Led Kafka rollout linking BMW's warranty-case management systems into a unified real-time event pipeline, replacing siloed stage-by-stage processing across the warranty admin lifecycle
- Technical Maintenance Manager for the resulting high-volume transactional system: production support, deployments, CI/CD, and database administration
- Mentored 3 Harambee Youth Employment Accelerator graduates through their 6-month rotation on the team

Software Developer, Discovery Limited · Jun 2019 – Jul 2021

- Designed reusable CRM components adopted across Discovery's product lines, reducing duplication for downstream teams
- Prototyped candidate architectures and presented in the Software Architect forum to shape company-wide technical standards

Software Developer, Hello Group · Jan 2016 – May 2019

- Championed Kubernetes and WSO2 adoption, establishing the standard microservices infrastructure across the company
- Delivered FinTech compliance systems audited by SARB and Deloitte UK
- Built remittance interfaces expanding the business into Malawi, Zimbabwe, and Bangladesh
- System lead for a critical project during a month-long deployment in India

OPEN-SOURCE & UPSTREAM CONTRIBUTIONS

17 commits merged across five projects, plus first review credit (Reviewed-by) on another contributor's regulatory analysis.

- **wpa_supplicant (hostap):** 7 commits merged Jul–Aug 2026 fixing Wi-Fi 7 multi-link (MLO) association — stale per-link BSS entries, ML-probe channel scanning, per-link key clearing, duplicated association-request IEs, and a PMKSA-caching deauthentication fallback — with two hwsim regression tests. Reproduced against real AP hardware; ships in 2.13.
- **Linux kernel (mainline):** RTL8922AU USB RF-calibration timeout fix, with 5 independent Tested-by tags across USB chipsets, plus a cfg80211 MLO group-key fix; a further cfg80211 patch under review.
- **NetworkManager:** Wi-Fi 7 MLO awareness for the daemon's background-scan logic (MR !2405, merged Jul 2026): parses the AP-MLD address from the Multi-Link element so a single-AP MLO association keeps the relaxed scan cadence, eliminating scan-induced traffic stalls. Landed after a five-revision review with the core maintainers.
- **wireless-regdb:** 7 commits across six jurisdictions (ZA, HK, RU, KR, UA, TG), each verified against primary regulatory instruments (ICASA, OFCA, MSIT, SCRF, ETSI). Obtained a written ruling from a national regulator (ARCEP Burkina Faso) on the applicable 6 GHz limits before submitting a patch.
- **morrownr/rtw89:** PR #76 backporting the kernel RF-calibration fix to the out-of-tree DKMS driver.

SELECTED PERSONAL PROJECTS

- **Endo Lab** (endolab.net): production research-literature search platform indexing ~360k papers from 10 upstream sources with per-record provenance. Hybrid retrieval — BM25 plus pgvector kNN fused with reciprocal rank over MedCPT domain embeddings (Postgres/ParadeDB, FastAPI, HNSW); self-hosted.

SKILLS

Primary stack: Angular (6 yrs), TypeScript, JavaScript, GraphQL, Highcharts, Storybook

Backend & platform: Java (9 yrs), Spring Boot (9 yrs), Kafka (2 yrs), Kubernetes, Docker, microservices, REST/GraphQL API design

Cloud & infra: Azure, AWS, GCP, GitHub Actions CI/CD, Prometheus, Grafana

Systems / low-level: Linux kernel modules, wireless stack, C (kernel-adjacent), bash

Practice: Distributed team leadership, mentoring, Agile/Scrum, code review, technical writing (kernel-grade commit messages)

EDUCATION & CERTIFICATIONS

- **Diploma in IT: Software Development:** Varsity College, 2013–2015 (Top Student, all 3 years)
- Professional Scrum Master (PSM 1), Professional Scrum Developer (PSD 1)
- ICF Coaching Skills for Managers (30 hours)